

Rohit Ramaprasad

San Jose, CA | (858) 214-0593 | r.rohit5557@gmail.com
<https://www.linkedin.com/in/rohit-ramaprasad> | <https://github.com/curiousgradient>

EDUCATION

Master of Science in Computer Science

Sep'22 – Dec'23

University of California San Diego (UCSD), La Jolla, CA, **GPA 4.0/4.0**

Bachelor of Engineering in Computer Science

Aug'18 – May'22

Birla Institute of Technology and Science Pilani (BITS Pilani), Pilani, India, **GPA 9.28/10.00**

INTERESTS

Generative AI, Reinforcement Learning, Computer Vision, Distributed Computing

WORK EXPERIENCE

Adobe, San Jose, CA

Machine Learning Engineer

Jan'24 – Present

- Led the design, implementation, evaluation, and rollout of an LLM-based re-ranking framework for citation generation used across all Adobe Acrobat GenAI features, improving relevance and coverage by 20% versus standard IR systems for single-document, multi-document, and multilingual workflows, strengthening trust in AI-generated responses.
- Delivered a fast, cost-efficient citation pipeline for a high-traffic summarization feature serving 10M requests per month, achieving high citation coverage and strong relevance without additional neural rerankers.
- Architected and built a robust ground-truth citation system to evaluate the production pipeline and independently measure retrieval and re-ranking quality, automatically identify failure stages, and drive targeted, data-informed improvements.
- Developed novel approaches for complex, synthesized, and table-based AI citations, resulting in two patent applications; received an Adobe-wide hackathon honorable mention for an innovative solution for tabular-data attribution.
- Served as a technical leader and subject-matter expert for attribution systems, mentoring engineers, leading design and code reviews, resolving high-priority production issues, and elevating team-wide debugging and evaluation practices.

Machine Learning Engineer Intern

Jun'23 – Sep'23

- Researched state-of-the-art LLM prompt injection techniques and built a repository of more than 500 inputs to bypass Acrobat GenAI safeguards
- Conducted an in-depth evaluation of heuristic-based prompt leakage mitigation techniques and derived actionable insights that improved safeguards by more than 30%
- Implemented a novel dual-LLM architecture using Langchain to detect prompt leakage, adhering to prompt engineering best practices. Rigorously evaluated false positive and false negative rates of different variations of the above architecture and improved over heuristic-based leakage detection by more than 10%

Playment (now TELUS International), Bangalore, India

Jan'22 – Jun'22

Machine Learning Engineer Intern

- Proposed, implemented, and pushed to production a novel image segmentation annotation approach comprising salient object detection and click-based interactive segmentation which decreased image annotation times by more than 60% while achieving an IoU of almost 95%
- Prototyped Flask backend hosting our ML model with image compression optimizations for sending large RGB images and receiving binary masks

Intelligent Multimodal Video Analysis Group, McGill University, Canada

May'21 – Jan'22

Research Software Development Intern with Prof. Dr. Martin Levine and Dr. Farzaneh Askari

- Innovated and implemented an attention-based convolutional recurrent neural network using PyTorch for detecting the “key” players involved in ice-hockey penalties. Used the Ax optimization tool for Bayesian hyperparameter tuning and MLflow for analyzing model performance. Trained model on SLURM cluster using distributed data-parallel strategy and ensuring reproducible results.
- Built an interactive GUI using OpenCV to be used by annotators for labelling and cleaning the dataset. Paper accepted at the CVsports Workshop, CVPR'22.

MultiCog Research Group, BITS Pilani, India

Jan'20 – Aug'20

Research Assistant with Assistant Prof. Dr. Pratik Narang

- Optimized RetinaNet for small object detection by tuning backbone depth, anchor box configuration, input image resolution and more.
- Implemented post-processing algorithms like soft-NMS and bounding-box voting to study the effect on mAP. Our model, Deep Optimized High-Resolution RetinaNet, beat the baseline by 6% mAP at the VisDrone Object Detection Workshop, ECCV'20 and we co-authored the challenge results paper.

ACADEMIC SERVICE & TEACHING

Invited Reviewer CVsports Workshop, CVPR'26

Mar'26

Reviewed papers for the CVsports Workshop, CVPR'26

Graduate Teaching Assistant – Scalable Data Systems

Mar'23—Jun'23, Sep'23—Dec'23

University of California San Diego (UCSD), La Jolla, CA

- Designed and implemented programming assignments and datasets for distributed parallel computing using Dask for Data Science graduate class Scalable Data Systems
- Conducted weekly office hours for 1-on-1 student guidance and doubt resolution
- Evaluated student assignments in Dask and PySpark on AWS EC2 servers

PUBLICATIONS

- 2024 Farzaneh Askari, Cyril Yared, **Rohit Ramaprasad**, Devin Garg, Anjun Hu, James J Clark, Video Interaction Recognition using an Attention Augmented Relational Network and Skeleton Data, **CVPR Workshops'24** [\[Paper\]](#)
- 2022 Farzaneh Askari, **Rohit Ramaprasad**, James J. Clark, Martin D. Levine, Interaction Classification with Key Actor Detection in Multi-Person Sports Videos, **CVsports Workshop, CVPR'22** [\[Paper\]](#) [\[Code\]](#)
- 2022 **Rohit Ramaprasad**, Sundaresan Raman, SEMFD-Net: A Stacked Ensemble for Multiple Foliar Disease Classification (ACM IKDD CODS COMAD Undergraduate Student Track'22) [\[Paper\]](#)
- 2021 Ayush Jain, **Rohit Ramaprasad**, Pratik Narang et al, AI-Enabled Object Detection in UAVs: Challenges, Design Choices, and Research Directions, **IEEE Network'21** [\[Paper\]](#)

2020 Dawei Du, **Rohit Ramaprasad**, Pratik Narang et al. VisDrone-DET2020: The Vision Meets Drone Object Detection in Image Challenge Results, **ECCV'20 Workshops** [\[Paper\]](#) [\[Code\]](#)

PATENTS

May 2025 U.S. Patent Application (filed; pending) — *Attributing Source Content to Generated Text Using Generative AI* (P14177-US-1), co-inventor

Nov 2024 U.S. Patent Application (filed; pending) — *Table Structure Attribution via Row-Column Description Contextualization* (P13655-US-1), co-inventor

AWARDS AND HONORS

- 2024 Earned an **Adobe-wide hackathon honorable mention** for developing an innovative approach to handling attributions involving creative computation on tabular data
- 2021 Awarded the distinguished **Mitacs Globalink Research Fellowship** to pursue a fully funded research internship at McGill University, Montreal, Canada
- 2019 Our team was awarded **the first prize at the National IBM Hack Challenge'19** for developing a web application that calculates the friend affinity between two users using their digital footprint [\[Code\]](#)
- 2018 Awarded the prestigious **KVPY Fellowship, ranking 403rd nationally** in India for demonstrated aptitude in scientific research through a rigorous selection process involving competitive examinations and interviews. Also secured an **All India Rank of 2,745 among over 200,000** qualified candidates in the highly competitive **JEE Advanced** examination, earning admission opportunities to several Indian Institutes of Technology (IITs) across the country.

SELECTED COURSEWORK

Graduate - Scalable Data Systems for ML, Statistical NLP, Advanced Computer Vision, Introduction to Robotics, Search and Optimization, Recommender Systems

Undergraduate - Machine Learning, Design and Analysis of Algorithms, Operating Systems, Object-Oriented Programming, Applied Statistical Methods, Computer Architecture, Fuzzy Logic, Optimization